Fixing GitHub API Rate Limits on de.NBI VMs

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Problem

Tools on the de.NBI VM fail with `API rate limit exceeded for <IP>`. Behind the error is how de.NBI cloud VMs reach the internet: they go out through a shared NAT IP. Unauthenticated GitHub requests are limited to **60 per hour per IP**, and every user behind that IP shares the same quota.

Fix: authenticate with a personal access token. The limit then becomes **5,000 per hour**, counted against your own account.

1. Create a token

1. On github.com, go to **Settings → Developer settings → Personal access tokens → Fine-grained tokens → Generate new token**.
2. If you only fetch public repos, select **Public repositories (read-only)**. Do not add any other permissions.
3. Copy the token, which starts with `ghp_` or `github_pat_`. Treat it like a password.

2. Set the token on the VM

Add the token to `~/.bashrc` so it persists across sessions:

```
echo 'export GITHUB_TOKEN=ghp_xxx' >> ~/.bashrc
echo 'export NXF_GITHUB_TOKEN=$GITHUB_TOKEN' >> ~/.bashrc   # for Nextflow
source ~/.bashrc
```

3. Check that it works

```
curl -s -H "Authorization: Bearer $GITHUB_TOKEN"
https://api.github.com/rate_limit | grep -A3 '"core"'
```

The output should show `"limit": 5000`. If it shows `60`, the token is not being picked up.

4. Make sure each tool uses the token

Nextflow / nf-core. Recent versions read the environment variable. For older versions, or to be safe, also create `~/.nextflow/scm`:

```
providers {
  github {
    user = 'your-github-username'
    password = 'ghp_xxx'
  }
}
```

gh CLI. Run `gh auth login`, or rely on `GITHUB_TOKEN`.

git clone and pip/conda installs from GitHub. Rewrite GitHub URLs so they include the token:

```
git config --global url."https://${GITHUB_TOKEN}@github.com/".insteadOf
"https://github.com/"
```

Your own scripts. Add the header `Authorization: Bearer $GITHUB_TOKEN` to requests to `api.github.com`.

Cluster jobs (SLURM etc.). Make sure the variable is exported to the jobs as well, for example in the job script or in the `env` block of the Nextflow config.

Troubleshooting

- **Still limited after setup:** run the check in step 3 and note the full error line. That shows which tool is still sending unauthenticated requests.
- **Token expired:** fine-grained tokens have an expiry date, so regenerate the token and update `~/.bashrc`.
- **Security:** never commit the token to a repository, and don't put it in shared config files on multi-user VMs.